

The Database Environment - Ch. 1

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1 The Database Environment

1.1 Defining a Database

People have come to think that anything that stores or manipulates is database software. Nonetheless, that's not the case. The idea behind a database is that user:

- Has no need to worry about the way in which data are physically stored on the disk.
- Phrases data manipulation requests in terms of data relationships.

A piece of software known as a *database managements system* (DBMS) then translates between the user's request for data and the physical data storage. In a database, the relationships are between those things mentioned earlier (the customers, order, inventory items, etc.) rather than between individual pieces of data.

1.2 Systems that Use Databases

There are two major types of systems that use databases in medium and large organizations:

Transaction Processing These systems handle day-to-day operations of an organization. Sales, accounting, manufacturing, human resources, etc.

Analytical Processing These systems are used in support of the analysis of organizational performance, making high-level operational decisions, and strategic planning.

1.3 Data Ownership

Who "owns" the data in your organization? The idea of data ownership has some important implications. Whether we have isolated databases where data is isolated to departments, or we have an IT department that carefully doles out permissions to access data, it must be clear that data should be organized in a way that balances security while still being accessible to the people who need to have access.

1.4 Database Software: DBMSs

There is a wide range of DBMS software available today. Azure, FileMaker, Helix, Oracle, etc. However, most are intended for enterprise use. There are versions of some enterprise software that are aimed at smaller business, but most is primarily intended for large installations using mainframes. Regardless of the DBMS you choose, there are some capabilities that you should expect to find:

- A DBMS must provide facilities for creating the structure of the database. Developers must be able to define the logical structure of the data being stored, including the relationships among data.
- A DBMS must provide some way to enter, modify, and delete data, i.e. languages like SQL.
- A DBMS must also provide a way to retrieve data, and support complex query statements using Boolean algebra (AND, OR, NOT operators), and should be able to perform at least basic calculations.
- A DBMS should provide methods for restricting access to data.